

HAIQA KHAN

Machine Learning Engineer | NLP | LLMs | Agentic AI

haiqakhan721@gmail.com | +92 332 3108852 | linkedin.com/in/haiqa-khannn | github.com/haiqakhan721 | Karachi, Pakistan

SUMMARY

Final-year Computer Science student at FAST-NUCES with production ML experience at Syslab, including a system deployed live at United Bank Limited, one of Pakistan's largest banks. Specializes in NLP and low-resource language AI, with hands-on work in LLM fine-tuning, multi-agent pipelines, and RAG systems. Focused on the gap between model training and real-world deployment.

EXPERIENCE

Machine Learning Intern | Syslab | Karachi, Pakistan *June 2024 - Present*

- Built an NLP-based **resume parsing system deployed in production at United Bank Limited**, automating structured extraction of candidate information and reducing manual CV processing time across the bank's recruitment pipeline.
- Designed a **document verification module** to authenticate user-submitted files, improving pipeline data integrity and reducing downstream processing errors.
- Contributed to the **MLOps pipeline**: model deployment, performance monitoring, and production tuning in a live environment used by real clients.
- Worked in an Agile team with senior engineers, participating in code reviews and sprint planning to deliver milestones on schedule.

PROJECTS

Urdu Ustad | Python, Transformers, PEFT/LoRA, RAG, FastAPI *2025 (FYP)*

- Fine-tuned a small language model on a custom Urdu curriculum dataset using **LoRA adapters**; integrated a RAG pipeline to ground answers in source material and reduce hallucinations on out-of-distribution queries.
- Built a full system pipeline covering Urdu text cleaning and tokenization, SLM inference, RAG retrieval, and a **video delivery API**, deployed end-to-end for real student use.

Whisper Urdu ASR | Python, Hugging Face, Whisper, Gradio, Common Voice *2025*

- Fine-tuned openai/whisper-small** on Mozilla Common Voice 17 (Urdu, ~220h) with custom data augmentation and YouTube-scraped conversational audio to improve real-world robustness beyond read-speech benchmarks.
- Evaluated using **Word Error Rate (WER)** and deployed as a live inference app on Hugging Face Spaces. Baseline Whisper-small scores >90% WER on Urdu; fine-tuning closes this gap substantially.

Agentic Clinical Coder | Python, LLMs, ChromaDB, Pydantic, Groq, ICPC-2 *2025*

- Designed a **3-agent A2A pipeline** (Extractor, RAG Retriever, Selector) to automate ICPC-2 medical coding from clinical notes, addressing the context-overload failure mode of single-prompt LLMs on small models.
- Benchmarked 4 LLMs** (Llama 3.1 8B, Llama 3 8B, Gemma2 9B, DeepSeek-R1-Distill) across 437 clinical queries; best model achieved **F1 0.748 with 100% format adherence**, a ~65% relative improvement over monolithic single-prompt baselines on identical model sizes.

RAG Pipeline | Python, FastAPI, ChromaDB, Docker, Hugging Face Spaces *2025*

- Built and Dockerized a **production-ready RAG API** with FastAPI, ChromaDB vector storage, local sentence-transformer embeddings (all-MiniLM-L6-v2), and Groq LLM for answer generation.
- Deployed to Hugging Face Spaces via Docker SDK; supports PDF/DOCX ingestion, full CRUD document management, and a Swagger UI interface for API testing.

EDUCATION

B.S. Computer Science*Expected June 2026*

National University of Computer and Emerging Sciences (FAST-NUCES), Karachi

Relevant Coursework: Machine Learning, Deep Learning, Artificial Intelligence, Data Structures & Algorithms, Computer Networks, Statistics & Applications

A Levels | 2As, 2Bs*June 2022*

Army Public School & Cambridge System, Karachi

SKILLS

Languages: Python, C++, JavaScript, SQL, HTML/CSS

ML & AI: Hugging Face Transformers, PEFT/LoRA, LangChain, ChromaDB, sentence-transformers, Groq, Ollama, Pydantic

Evaluation: WER, F1, RAGAS, BERTScore, prompt engineering, structured outputs

Infrastructure: FastAPI, Docker, Git, AWS, Jupyter, Hugging Face Spaces

Core Areas: NLP, LLM fine-tuning, RAG pipelines, agentic AI systems, low-resource language modeling, MLOps